Physics 5C Discussion (Week 6)

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Problem 1 Human body is a huge conductor with finite resistance. Across Jinyan's two dry hands, the resistance is around $R = 100\,\mathrm{k}\Omega$, but an electric current as low as $90\,\mathrm{mA}$ can cause ventricular fibrillation.

Part (a) The voltage supplied by a standard U.S. power outlet is $120\,\mathrm{V}$. For doing a physics demo, will it be too risky for Jinyan to stick his hands into the power outlet? [1]

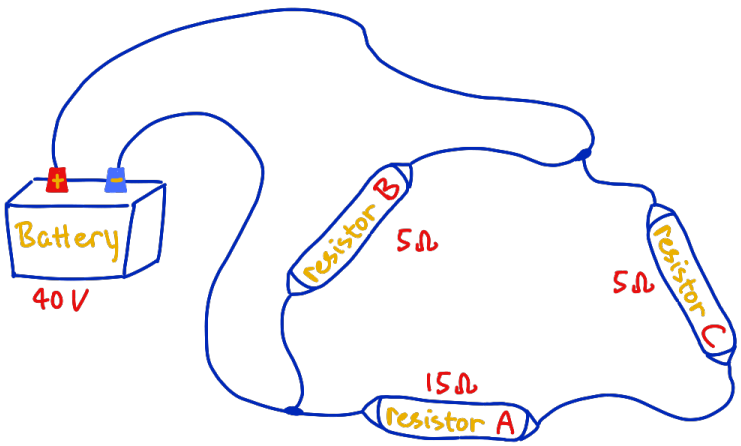
Part (b) It takes around $20\,\mathrm{kJ}$ of energy to cook an egg. The power (energy per second) delivered by a current through a conductor is calculated via

$$P = IV\,.$$

Hypothetically, if Jinyan sticks his hands into the standard power outlet for 10 seconds, will he be cooked? [2]

The followings are standard circuit problems. I will show you the full process to solve Problem 2 during discussion, and Problems 3 and 4 are left to you for practice. For these problems, assume that the resistances in the battery and wires are negligible.

Problem 2 Jinyan has built a circuit as shown, but he has trouble analyzing the circuit. Please help him out!



Part (a) Based on the wiring of the components, draw the equivalent circuit diagram.
Part (b) Find the voltage drop across resistor A.
Part (c) How should I use a voltmeter to measure the voltage across resistor A?

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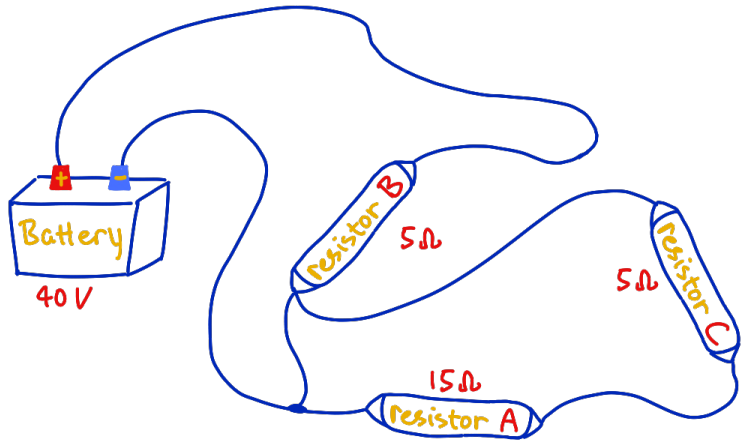
[1] Don't be crazy, please don't do it :(

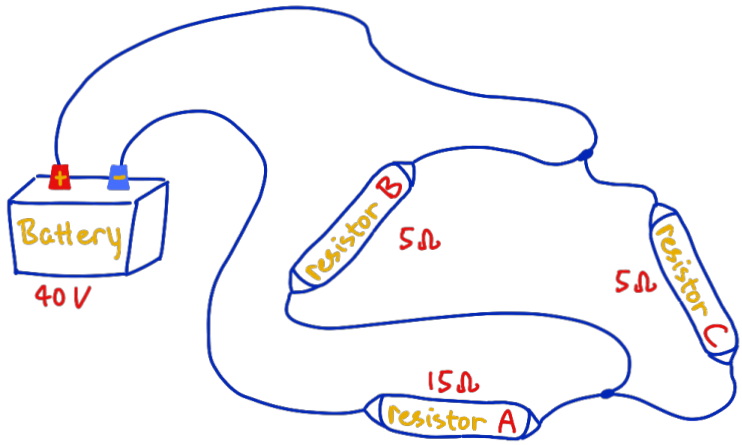
[2] Again, please don't do it.

Problem 3 Jinyan's evil labmate Michael sneakily changes one of the wires.



Part (a) Based on the wiring of the components, draw the equivalent circuit diagram.
Part (b) Find the current through resistor A and resistor B.[3]
Part (c) How should I use an ammeter to measure the current through resistor A?

Problem 4 Jinyan finds something weird going on and rewired the components, but he gets confused again. Please help!



Part (a) Based on the wiring of the components, draw the equivalent circuit diagram.
Part (b) Find the total equivalent resistance of the circuit.
Part (c) Find the current through the battery.
Part (d) Find the current through resistor A.
Part (e) Find the voltage drop across resistor A.
Part (f) Find the voltage drop across resistor B.
Part (g) Find the current through resistor B.
Part (h) Find the power dissipated by resistor B.

[3]Other than the quantities listed on the figure, you should NOT reuse anything you computed in Problem 2. These are completely different circuits. If you draw out the correct circuit diagram and think about it carefully, you will find Problem 3 is very straightforward and easy, and you should be able to answer all parts of Problem 3 in 1 minute.